

M1_Requirements_SignOff_P7

M1 — Requirements Sign-off

Project P7: Smart Hospital Dashboards (PRD-08)

Team: Greeshma & Sudhanva Holla **Date:** 11 July 2026 **Scope:** Phase-1 / MVP, S-tier (●) requirements only

1. Scoped Requirement List

Only the following Phase-1, S-tier requirements from PRD-08 are in scope for this sprint:

FR	Requirement	Build Spec (what we will actually build)
FR-3	Role-based dashboards with unit/department scoping and drill-down permission gates	A dashboard view that changes based on logged-in role (e.g. Dept Head sees only their ward; Superintendent sees all). Clicking a tile drills into row-level detail <i>only</i> if the role is permitted.
FR-5	Statutory report generator: HMIS monthly format, review-and-submit workflow	A button that compiles seeded event data into an HMIS-style monthly report (fixed columns/format), shows a preview screen, and requires a manual “Submit” click before it’s marked final.
FR-7	Scheduled digests (daily/weekly) per role via email/WhatsApp	A script that generates a plain-text/HTML digest (census, admissions, discharges, flags) and either emails it or logs a mock “WhatsApp send” (stub, since no real gateway).
FR-10	De-identification layer: de-identified default views, small-cell suppression	Patient names/IDs replaced with pseudonymised tokens by default on all dashboard/wall views. A toggle can reveal identified data only for permitted roles. Any breakdown with <5 records in a cell is suppressed/shown as “—”.
FR-14	Audit of dashboard access & exports (who saw/exported what)	Every dashboard view and every report export writes a row to an <code>audit_log</code> table: user, timestamp, what was viewed/exported.

Explicitly out of scope for M1-M4: Command centre, alerting engine, NABH QI module, self-service explorer, surge feeds, benchmarking, AI features (all Phase 2/3 or M/L tier).

2. User Story Shortlist (≤6)

1. As a **department head**, I want my dashboard filtered to my unit with drill-down to (permission-gated) patient level so I can act, not just observe. (FR-3)
2. As an **MRD/statistics officer**, I want monthly HMIS reports generated from the warehouse with a review-and-submit step so compilation effort drops to minutes. (FR-5)
3. As an **owner (S tier)**, I want a daily digest — census, admissions, discharges, flags — on email/WhatsApp so I don’t have to log in to stay informed. (FR-7)
4. As a **DPO**, I want certainty that wall displays and exports show de-identified data by default, with identified drill-down only for authorised roles. (FR-10)
5. As an **admin/compliance officer**, I want an audit trail of who viewed or exported what data, so we can prove accountability if questioned. (FR-14)
6. As a **medical superintendent**, I want a single top-level view across departments (even if basic) so I can spot which unit needs attention. (light FR-3 extension)

3. Data Model Draft (ERD — Star Schema)

Fact table: `FACT_EVENTS` | Column | Type | Notes | |—|—|—| | `event_id` | INTEGER PK | | | `date_key` | INTEGER FK | → `DIM_DATE` | | `ward_key` | INTEGER FK | → `DIM_WARD` | | `payer_key` | INTEGER FK | → `DIM_PAYER` | | `patient_pseudo_id` | TEXT | de-identified token, not real UHID | | `event_type` | TEXT | admission / discharge / transfer / ED_registration / doctor_consult | | `event_timestamp` | DATETIME | | | `admission_duration` | INTEGER | length of stay in days, populated on discharge events | | `door_to_doctor_minutes` | INTEGER | populated on doctor_consult events | | `occupied_beds` | INTEGER | ward occupancy snapshot at event time | | `role_scope` | TEXT | which role(s) can see this row identified |

Dimension tables: - `DIM_DATE` (`date_key` PK, `full_date`, `day`, `month`, `year`, `weekday`) - `DIM_WARD` (`ward_key` PK, `ward_name`, `department`, `total_beds`) - `DIM_PAYER` (`payer_key` PK, `payer_name`, `payer_type`) - `DIM_USER` (`user_key` PK, `username`, `role`, `department`) — drives FR-3 & FR-10 gating; replaces a separate role-dimension table by folding role/department directly onto the user

Relationships: `FACT_EVENTS` uses `DIM_DATE`, occurs_in `DIM_WARD`, billed_to `DIM_PAYER`, viewed_by `DIM_USER` — each a many-to-one from fact to dimension.

Support tables (not fact/dim, but needed for FRs): - `audit_log` (`log_id`, `user_id`, `role`, `action`, `target_view_or_report`, `timestamp`) — FR-14 - `reports_generated` (`report_id`, `report_type`, `period`, `status[draft/submitted]`, `generated_at`, `submitted_by`) — FR-5 - `digest_log` (`digest_id`, `role`, `channel[email/whatsapp_stub]`, `sent_at`, `content_snapshot`) — FR-7

KPI Definitions (formulas): - **Occupancy %** = (`occupied_beds` ÷ `total_beds`) × 100 — `occupied_beds` now lives directly on `FACT_EVENTS` as a per-event snapshot, joined to `DIM_WARD.total_beds` - **ALOS (Average Length of Stay)** = average of `admission_duration` across discharge events in the period (pre-aggregated at event time rather than computed by subtracting timestamps downstream) - **Door-to-doctor time** = average of `door_to_doctor_minutes` across doctor_consult events in the period (same pattern — captured on the event, not derived later)

Design note vs. the M1 draft: the earlier draft computed ALOS and door-to-doctor by subtracting timestamps at query time. This revision pushes that calculation upstream — the source system (or ingestion pipeline) writes the duration directly onto the event row. This trades a small amount of ingestion complexity for much simpler, faster dashboard queries (no timestamp math needed on every tile refresh).

4. Compliance Checklist (from PRD-08 §8 Regulatory)

Instrument	Obligation	Feature it forces
DPDP Act 2023 + Rules 2025	Purpose limitation; de-identified by default; access logging; data minimisation in exports	FR-10 (de-identification), FR-14 (audit log)
DPDP Act 2023	Breach-notification readiness	FR-14 (audit trail is first evidence source in a breach review)
HMIS/NHM reporting formats	Faithful mapping to current HMIS indicators	FR-5 (report generator uses HMIS column format)
CERT-In directions 2022	Log retention (180 days), India-time NTP sync	<code>audit_log</code> retention policy note (even if demo only keeps short window, document the requirement)
ABDM / EHR Standards 2016	Coding standards for comparable metrics	Noted as N/A for MVP demo (no real ABDM integration; stub only)

Viva prep — “which rule forces this feature?”: - De-identified default view → **DPDP purpose limitation + data minimisation** - Audit log on every view/export → **DPDP access-logging obligation + CERT-In log retention** - Review-and-submit step on reports (not auto-submit) → **DPDP + HMIS accountability** — a human must own what’s submitted to a statutory body - Small-cell suppression → **DPDP re-identification risk control**

Data source decision (needed before M2): Coordinate with a P1 (Patient Registration) or P2 (Bed Management) team for a live event feed if possible; otherwise commit to the provided sample CSVs. *(Decide and note here before Sunday’s design freeze.)*

Notes for M2 (due Sun 12 Jul — don’t forget)

- Finalize ERD + SQLite DDL
- API/route list
- 3–5 screen wireframes (2 dashboards min per P7 table)
- Seeded sample dataset

- Walking-skeleton repo (app boots, 1 end-to-end flow stubbed)